

A Scalar Counterexample to a Uniform Property (S) Modulus Question

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns Open Problem (3) from:

William B. Johnson and Tomasz Kania, *Uniform Property (S)*, arXiv:2602.09106.

The source location is Section 5, Open problems, page 19 of the source PDF; in the parsed source this is line 1071 of `main.tex`.

5. OPEN PROBLEMS

Let us list open problems concerning uniform property (S).

- (1) Does c_0 have equivalent renormings with uniform property (S)?
- (2) Does every space have an equivalent renorming with uniform property (S)?
- (3) Is $U(d, a)$ always smaller than da (up to a constant)?
- (4) Characterise those pointed metric spaces whose Lipschitz-free Banach spaces have uniform property (S). For example, by Rademacher's theorem, $F([0, 1])$ is isometric to $L_1[0, 1]$ so it does have uniform property (S), unlike $F(\mathbb{N})$ that is isometric to ℓ_1 .
- (5) If X has uniform property (S), must X^{**} have it too? The converse follows from the principle of local reflexivity.

Problem (3) asks:

Is $U(d, a)$ always smaller than da (up to a constant)?

We give a negative answer to the statement as written, in the real Banach-space category, even if one restricts to spaces with uniform property (S).

Definitions

For a normed space X , Johnson and Kania define

$$U_X(x, y; a) = \sup_{\|z\| \leq a} | \|x + z\| - \|y + z\| |$$

for $x, y \in S_X$ and $a > 0$, and

$$U_X(d; a) = \inf_{\substack{x, y \in S_X \\ \|x - y\| \geq d}} U_X(x, y; a).$$

The open question asks whether there is a universal constant C such that

$$U_X(d; a) \leq Cda$$

for all relevant spaces X , all $0 < d < 2$, and all $a > 0$.

Counterexample

Theorem 1. *There is no universal constant C such that*

$$U_X(d; a) \leq Cda$$

for every real Banach space X with uniform property (S), every $0 < d < 2$, and every $a > 0$.

Proof. Let $X = \mathbb{R}$ with its usual norm. Then

$$S_X = \{-1, 1\}.$$

Consequently, if $0 < d < 2$, the only admissible ordered pairs $(x, y) \in S_X \times S_X$ with $\|x - y\| \geq d$ are $(1, -1)$ and $(-1, 1)$. Therefore

$$U_{\mathbb{R}}(d; a) = U_{\mathbb{R}}(1, -1; a).$$

We compute this quantity. For $z \in \mathbb{R}$,

$$||1 + z| - |-1 + z|| \leq 2$$

by the reverse triangle inequality. If $0 < a \leq 1$, then for $z \in [-a, a]$ one has $1 + z \geq 0$ and $1 - z \geq 0$, so

$$|1 + z| - |-1 + z| = (1 + z) - (1 - z) = 2z.$$

Taking absolute values and then the supremum over $|z| \leq a$ gives

$$U_{\mathbb{R}}(d; a) = 2a \quad (0 < d < 2, 0 < a \leq 1).$$

In fact the full scalar formula is

$$U_{\mathbb{R}}(d; a) = \min\{2a, 2\}, \quad 0 < d < 2, a > 0.$$

In particular, $U_{\mathbb{R}}(d; a) > 0$ for every $0 < d < 2$ and every $a > 0$, so $\mathbb{R}$ has uniform property (S).

Now suppose that a universal constant C existed. Applying the proposed bound to $X = \mathbb{R}$ and $0 < a \leq 1$ would give

$$2a = U_{\mathbb{R}}(d; a) \leq Cda$$

for every $0 < d < 2$. Since $a > 0$, this implies

$$2 \leq Cd$$

for every $0 < d < 2$, which is impossible as d tends to zero. □

Verification Notes

The proof is exact. The script `code/check_scalar_modulus.py` numerically compares the formula

$$U_{\mathbb{R}}(d; a) = \min\{2a, 2\}$$

against a fine grid maximization of

$$||1 + z| - |-1 + z|| \quad (|z| \leq a)$$

for several values of a . This is only a sanity check; the scalar calculation above is the proof.

Limitations and Human Review Guidance

This is a literal counterexample to Problem (3) as printed. It is also a small edge case: the obstruction is that the real one-dimensional unit sphere has no pairs at distance d for $0 < d < 2$ except the antipodal pair, whose distance is 2. Thus the infimum defining $U(d; a)$ cannot probe genuinely small-distance geometry.

The result should therefore be read as identifying a missing hypothesis in the printed question. A more substantive modified problem might ask for the same upper bound in infinite-dimensional spaces, in spaces with connected unit sphere, or under a condition guaranteeing admissible pairs at every small distance. This packet does not resolve those strengthened versions.

Novelty Check

No external literature search beyond the source paper and the local pipeline corpus was performed. The argument is elementary enough that it may be known informally; the claim here is only that it answers the printed statement exactly.